



Introduction of MongoDB



MongoDB is a NoSQL (non-relational) database.
It stores data in a type of JSON format called **BSON**.

A DATABASE

is a collection of data stored in a format
that can easily be accessed.



What is Database ?

Name

Age

Gender

☐ Male ☐ Female

Submit

A simple form collects information from a user — their name, age, and gender.



What is Database ?

Name

Age

Gender

☐ Male ☐ Female

Submit

Collection of Data

Name	Age	Gender
Ram Kumar	21	Male
Salman Khan	22	Male
Meera Khan	21	Female
Sarita Kumari	21	Female
Anil Kapoor	22	Male

DATABASE

Multiple form submissions collected together become a database.



Type of Databases

SQL Databases

Name	Age	Gender
Ram Kumar	21	Male
Salman Khan	22	Male
Meera Khan	21	Female
Sarita Kumari	21	Female
Anil Kapoor	22	Male

Table

Oracle, MySQL, PostgreSQL, MS SQL Server

NoSQL Databases

```
{  
  name : "Rizwan"  
  age : 21,  
  gender : "male"  
}
```

```
{  
  name : "Salman"  
  age : 22,  
  gender : "male"  
}
```

Collection



Type of Databases

SQL Databases

Name	Age	Gender
Ram Kumar	21	Male
Salman Khan	22	Male
Meera Khan	21	Female
Sarita Kumari	21	Female
Anil Kapoor	22	Male

Table

Oracle, MySQL, PostgreSQL, MS SQL Server

MongoDB

NoSQL Databases

Document



```
{
  name : "Ram Kumar"
  age : 21,
  gender : "male"
}
```

Document



```
{
  name : "Salman"
  age : 22,
  gender : "male"
}
```

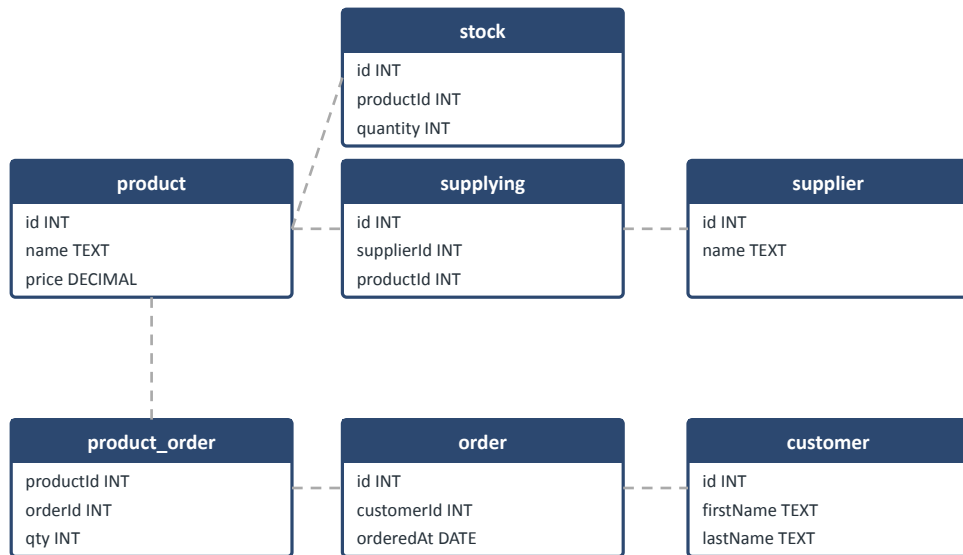
Collection



Type of Databases

SQL Databases

Relational Database



A relational database links multiple tables together through shared keys.



Difference between JSON and BSON

JSON

Data Types

BSON

Binary JSON

- String
- Number
- Boolean
- Array
- Object
- Null



Difference between JSON and BSON

JSON

Data Types

- String
- Number
- Boolean
- Array
- Object
- Null

BSON

Binary JSON

```
{
  "name" : "Ali",
  "age" : 25,
  "married" : true,
  "hobbies" : ["music", "sports"],
  "vehicle" : {
    "type" : "car", "vname" : "swift"
  }
}
```




Difference between JSON and BSON

JSON

Data Types

- String
- Number
- Boolean
- Array
- Object
- Null

BSON

Binary JSON

- String
- Double
- 32-bit Integer
- 64-bit Integer
- Boolean
- Array
- Object
- Null
- Regular Expression
- Timestamp
- Date
- ObjectId



Difference between JSON and BSON

JSON

BSON

Binary JSON

Format	Text-based (human-readable)	Binary (not human-readable)
Size	Larger due to text overhead	More compact
Speed	Slower to parse and write	Faster to parse and write
Usage	Web APIs and data exchange	MongoDB and database storage with a rich set of data types.



Benefits of MongoDB Database

- Flexible schema design.
- Horizontal scalability through sharding.
- High performance for read and write operations.
- Powerful query language.
- Built-in replication for high availability.
- Geospatial data support.
- JSON-like BSON data format.
- Real-time analytics capability.



What we will learn in this course ?

- Create Collection & Documents in database.
- Add Data in Collection.
- Update Data in Collection.
- Read data from Collection.
- Delete Data from Collection.
- Complex Search on Data.



CRUD Operations

Create · Read · Update · Delete



CRUD

CREATE

Add new documents to a collection.

READ

Retrieve existing documents from a collection.

UPDATE

Modify fields in existing documents.

DELETE

Remove documents from a collection.



CRUD Operation Methods

CREATE

```
insertOne()  
insertMany()
```

READ

```
find()  
findOne()
```

UPDATE

```
updateOne()  
updateMany()
```

DELETE

```
deleteOne()  
deleteMany()
```



Example: Nested Document Structure

- maker
- model
- fuel type
- transmission
- engine
 - cc
 - torque
- features
 - sunroof
 - airbags